

FINAL SEMINAR · OCTOBER 2026

Multi-Page Visually Rich Document Understanding

A survey of system design and an empirical attribution of failure

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Seminar Overview

01 Background

Visually rich documents, the single-page setting, and what changes once a document runs to many pages.

02 Survey of Architectures and Techniques

MP-VRDU organised through evidence management: a taxonomy of four architectural families, multimodal representation, and inference-time and training strategies.

03 Empirical Attribution of Failure Modes

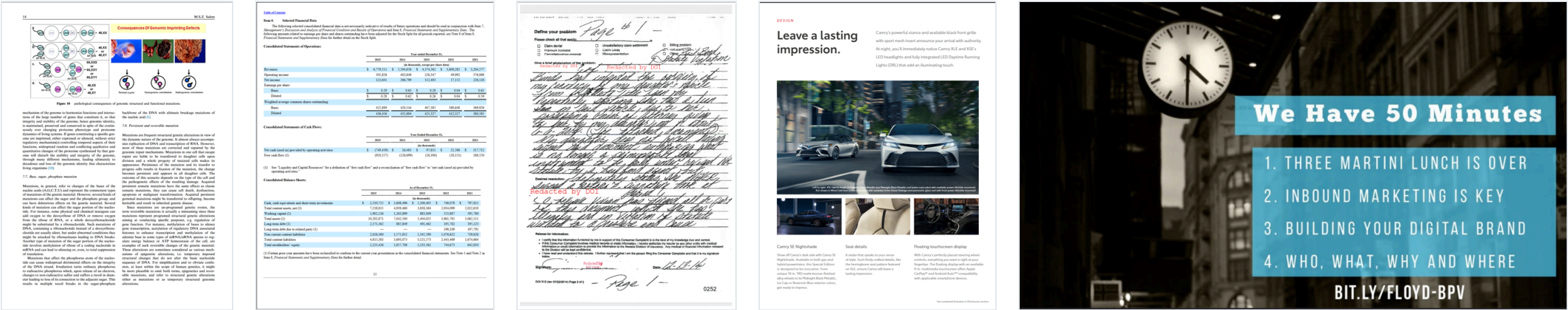
Failure attributed to three loci, representation, selection, and reasoning, through controlled interventions on a single pipeline.

04 Conclusion

A simple adaptive-trajectory pipeline built from those findings, and the open challenges that define the frontier.

Visually Rich Documents

A visually rich document is one whose meaning is carried jointly by text, layout, and visual content rather than by text alone.



Representative pages from MMLongBench-Doc

Text is small and dense

Typeset or handwritten, and legible only at far higher input resolution than a typical vision-language task requires.

Layout contains semantic information

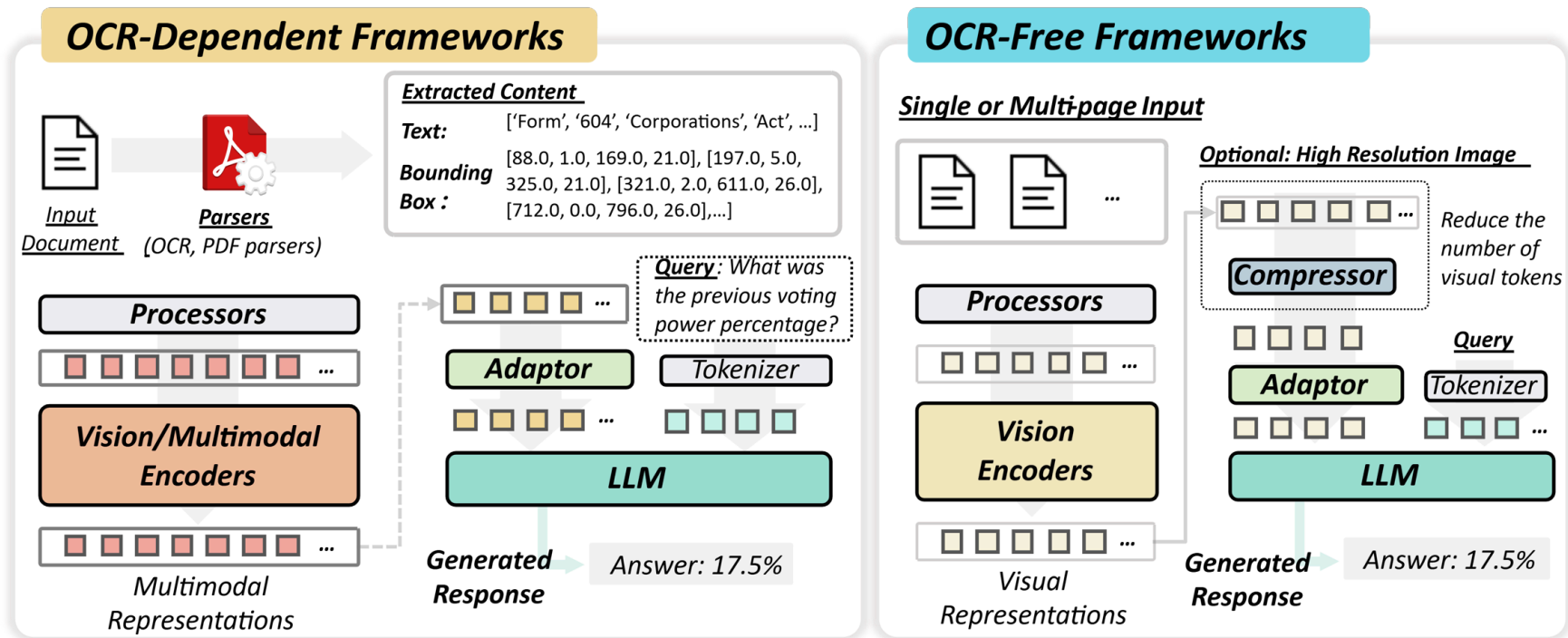
A number means one thing in a table's first column and another in its second, so position must be interpreted rather than read.

Reading order must be recovered

A page is a two-dimensional arrangement rather than a linear stream, so the order cannot be assumed.

Single-Page VRDU Overview

The single-page setting is largely solved, and multi-page systems inherit their per-page machinery from it.



OCR-dependent and OCR-free architectures for MLLM-based VRDU. Adapted from Ding et al.

OCR-dependent

Parse the page first, recovering text with its bounding-box coordinates. Captures structure without heavy visual pretraining, but ties accuracy to the parser's.

OCR-free

A vision encoder reads the rendered page from pixels. Handles handwriting and irregular layouts, but needs high resolution, so compression usually follows.

Challenges of the Multi-Page Setting

- 1 The document exceeds the context window**

Long multi-page documents routinely outrun what a backbone can hold, so the system must select which parts of the document reach the MLLM at all.
- 2 Scaling the context does not resolve it**

Long-context models attend poorly to evidence buried mid-input, and memory and compute grow with sequence length, so the achievable context is bounded by hardware.
- 3 Every required evidence unit must be located**

Answering often means integrating evidence from several parts of the document at once, and missing any one of them is enough to answer incorrectly or not at all.
- 4 Cross-page structure is not recoverable in isolation**

Section hierarchy and cross-references cannot be read off any single part of the document, yet questions often need them to gather the evidence in the first place.

Thesis by Compilation

This thesis is structured as a compilation of two papers written during this Honours year, each addressing one of its two aims.

1 **Managing Evidence at Document Scale: A Survey of Multi-Page Visually Rich Document Understanding**

Lewei Xu, Yihao Ding, Zihan Xu, Siwen Luo, Daochang Liu, Wei Liu

Main Conference of EMNLP 2026 · Chapter 2

AIM 1 Comprehensively survey techniques and architectures for multi-page document understanding, treating it as a problem distinct from the single-page setting.

2 **Locating Failure in Multi-Page Visually Rich Document Understanding: An Empirical Attribution**

Lewei Xu, Yihao Ding, Zihan Xu, Daniel Yitian Su, Daochang Liu, Siwen Luo, Yifan Peng, Wei Liu

Submitted to EACL, under review · Chapter 3

AIM 2 Attribute MP-VRDU failure to its underlying causes, and resolve the field's competing and largely untested claims about how these systems should be built.